

can be attached to mobile stands or maneuverable boom arms mounted on a wall or ceiling for easy positioning. Moreover, fixtures can incorporate panels of different sizes and quantities to control light output and coverage area.

Another advantage of LED surgical lights is that, unlike conventional lighting devices these do not radiate heat on the surgical area, which imparts an extra measure of comfort to the surgical staff and reduces air-cooling requirements of the area, thus creating extra energy savings.

Further, low emissions of IR and UV radiations from LEDs also make these lights safe for highly-controlled medical environments as these do not interfere with sensitive chemical compounds, diagnostic sensors or specialised treatments. In situations where more localised light is required at the surgical site, LED-based surgical headlights can be used. Such lights use a single high-output LED with high colour temperature to create a powerful beam of white light with a tight lighting field.

Such lights can also be directed to specific areas to compensate for any shadows in the operational area, thus providing more visibility at depth. Super-localised lighting inside the wound area and inside the body are primarily achieved through endoscopic devices that use a flexible or rigid fibre-optic system to transmit light to the inserted tip for internal illumination.

LED endoscopic lighting. Advanced solid-state (that is, LED) endoscopic lighting integrates LED modules on the head of devices in place of fibre optics. High-powered white LEDs are being used as the source of light for these devices because of their high luminous output, dynamic control and superior energy efficiency compared to traditional light sources. This puts LED sources of light directly at the point of illumination to ensure optimum light quality and emission.

For improved illumination, integrated designs and attachable LED lighting systems are being applied to such medical instrumentation as



Phototherapy in the medical field (Credit: <http://web.tradekorea.com>)

examination probes, tissue retractors and surgical forceps. One recent innovation in this direction is the development of a stretchable sheet of tiny micro-LED arrays measuring 100µm connected across an interlaced circuit mesh, transferred to a flexible polymer sheet and encapsulated in transparent rubber. The sheet is bio-compatible and can be attached to a number of surfaces and materials. **Flexible micro-LED arrays could lead to wearable light sources and high-tech surgical gloves.**

Treatment with light. In the medical field, phototherapy provides controlled exposure to light using lights of particular wavelengths (specific colour ranges) based on light intensity and duration of exposure to treat a variety of medical conditions and sleep disorders to promote general health and well-being. LEDs are also affected by the applied wavelength's ability to penetrate the skin in non-vision-related treatments. In fact, LEDs are becoming the primary light source for such applications because of their characteristics like high luminous output, energy efficiency and durability.

Arrays of high-powered LEDs or multi-chip modules can be easily integrated into a variety of treatment devices that are used to provide intense uniform light fields with controlled intensity. LED-assisted treatment devices can be portable, handheld applicators that provide a spot beam for small treatment areas, medium-sized lighting panels for partial body coverage or large-scale systems containing thousands of LEDs that can be used for complete body immersion. Such light arrays can also be populated with multi-coloured LEDs, which allow a single device to

use a specific wavelength for a desired treatment.

- Red-coloured light having the deepest tissue penetration in the visible light range can stimulate natural biological processes that promote healing of damaged and weakened cells.
- Blue-light phototherapy is used to treat neonatal jaundice in newborns.
- Disorders involving disturbance of the sleep cycle (circadian rhythms) are a common

experience for late-shift workers, jet-lagged travellers and those living in colder northern climates suffering from seasonal associative disorders. Moreover, lethargic symptoms of such people can be treated by timed/controlled exposure through the retina of the eye to bright white light at specific time periods. The light triggers a biochemical reaction in the brain, inhibiting the production of melatonin, which determines the sleep cycle.

- High-powered white LED arrays, which emit up to 10,000 lux, are being used to simulate the full spectrum brilliance of daylight sun.

Photodynamic therapy. Photodynamic therapy uses LED light to activate photo-sensitive drugs to treat such skin conditions as severe acne and cancerous skin lesions. The basic treatment consists of applying the dormant drug to the affected area of the skin and then exposing it to a specific wavelength of light, which causes the drug to undergo a chemical reaction and treat the targeted tissue.

Light sources typically consist of arrays of high-output LEDs, which can be used in small lighting instruments for spot treatment or moderate-sized lighting panels for larger treatment areas. **LEDs are ideal for photodynamic therapy applications because of their ability to emit a consistent, uniform band of narrow wavelength to which applied drugs are closely matched.**

Future LED medical applications for healthcare enhancement

By combining LED technology and medical research, several integrations have already been revealed, while others are under progress, showing